

PACOS

Personal AI Career Operating System

A lightweight multi-agent pipeline that turns a six-month manual job search into a two-week build, generating personalised outreach, reading your inbox for signals, and keeping a live tracker.

- Hard constraint: nothing auto-sends. You approve every action.

SCOPE

AI / ML / GenAI roles

REGION

Bengaluru · Pune · Hyderabad · GGN

STACK

Python · Claude · Gmail · CSV

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WHY PACOS EXISTS

The Idea

Job searching is high-repetition, low-feedback work, the exact kind of task that collapses under low motivation. PACOS replaces the manual grind with a small set of scripts that generate personalised outreach from a spreadsheet, read your inbox for job alerts and recruiter replies, and keep a single live tracker so nothing slips.

The bet: one strong, personalised touch beats fifty copy-pasted applications. PACOS makes the strong touch cheap to produce at volume, and keeps a human in the loop for every send.

What it replaces

The manual pain	What PACOS does instead
Hand-writing 15 cover letters	One Claude API call generates four assets per lead
Forgetting to follow up	Inbox monitor flags unanswered threads automatically
Losing track of applications	Tracker CSV updates after every action
Missing LinkedIn / Naukri alerts	Gmail reader parses notification emails daily
Wrong tone for the company	Domain classifier adapts startup vs enterprise voice

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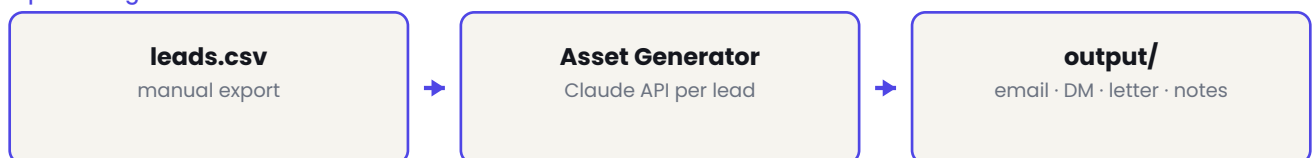
THREE LIGHT LAYERS

Architecture

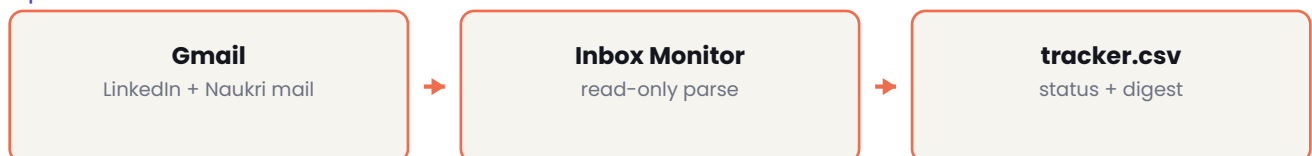
PACOS v1 is three small Python components and one Claude API call per lead. No orchestration framework, no database, no dashboard. Each layer runs independently and writes to plain files you can open in Excel or a text editor.

Layer	Component	What it does	Input / Output
1	Asset Generator	Reads leads, calls Claude once per lead, writes four outreach files	leads.csv » output/ folder
2	Inbox Monitor	Reads Gmail daily, parses LinkedIn + Naukri alerts, flags recruiter replies	Gmail (read-only) » tracker.csv
3	Tracker / CRM	One CSV logging every lead: status, dates, last action, next follow-up	all layers » tracker.csv

Pipeline A: generate



Pipeline B: listen



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MINIMAL & FREE

Tech Stack

Every dependency is free, open-source, or an API you already need for the search itself. The only paid item is Claude API credits, pennies at this scale.

Concern	Tool	Why this one
Asset generation	Claude API (sonnet-4-6)	Best personalisation quality; cheap at 500 leads
Scripting	Python 3.10+	Minimal setup, runs anywhere
Data handling	pandas	Read leads, write tracker, zero overhead
Inbox access	Gmail API (official)	Free, reliable, no scraping risk
Email parsing	email + BeautifulSoup	Parse LinkedIn / Naukri HTML mail
Scheduling	cron / Task Scheduler	No extra dependency
Storage	Local CSV files	No database for v1
Secrets	.env + python-dotenv	Keep API keys out of code

Deliberately not included: Sales Navigator scrapers, Proxycurl, Playwright, Apollo.io, Hunter.io, LangGraph, Supabase, and the React dashboard. Each adds cost, fragility, or platform-policy risk without earning its place in v1.

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SINGLE SOURCE OF INPUT

The Lead Sheet

The whole pipeline runs off one CSV. Export it free from a LinkedIn search, copy from Naukri, or type it by hand. No scraping tool required. Required columns are marked; the rest are optional enrichments that sharpen personalisation.

Column	Req	Source	Notes
full_name	Yes	LinkedIn / manual	
first_name	Yes	parsed	auto-derived by script
job_title	Yes	LinkedIn / Naukri	
company_name	Yes	LinkedIn / Naukri	
city	Yes	LinkedIn / Naukri	BLR · Pune · HYD · GGN
linkedin_url	Yes	LinkedIn	used in DM
persona_tag	Yes	manual / script	hr · engineer · manager · cto · ceo
email	No	manual lookup	if blank, DM-only outreach
company_size	No	LinkedIn	feeds domain classifier
industry	No	LinkedIn	feeds tone selection
company_website	No	manual / Google	cover-letter hook
funding_stage	No	Crunchbase / manual	startup vs scaleup
personalization_line	Auto	Claude API	filled by generator
domain_tag	Auto	script	startup · scaleup · enterprise · research

Getting leads without scraping

- LinkedIn free search: filter by title + city, note names and profile URLs
- Naukri recruiter profiles: visible without premium for most roles
- Company career pages: hiring-manager name from the About page or team list
- GitHub & conference speaker lists: ideal for research-oriented roles
- The job post itself: often names the hiring manager or team lead

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ONE CALL, FOUR FILES

Per-Lead Assets

For each row, the generator makes a single Claude API call and writes a dedicated folder of four ready-to-send files.

```
output/  
└─ {company_name}_{first_name}/  
   └─ cold_email.txt  
   └─ cold_dm.txt  
   └─ cover_letter.txt  
   └─ cv_notes.txt
```

Asset	Length	Purpose	Driven by
cold_email.txt	5–7 lines	Primary outreach when email is known	name · company · role · website · hook
cold_dm.txt	3–4 lines	LinkedIn DM, main channel without email	name · role · company · persona
cover_letter.txt	150–200 wd	Attached to formal applications	company · role · domain · hiring signal
cv_notes.txt	bullets	How to tweak your CV for this role	role · industry · domain · size

Domain-aware tone

domain_tag	Example companies	Tone applied
startup	Seed / Series A, <50 people	Casual, founder-direct, energetic
scaleup	Series B/C, 50–500	Growth-focused, impact & metrics
enterprise	500+, MNCs	Formal, process-aware, structured
product	SaaS / B2B product	Product thinking, user metrics
services	IT services, consulting	Delivery-focused, client-facing
research	Labs, academia, R&D;	Technical depth, rigor, publications

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CLOSING THE LOOP

Inbox Monitor

The monitor connects to Gmail through the official API (OAuth, read-only) and runs daily. It parses LinkedIn and Naukri alert emails and flags any direct recruiter reply, turning your inbox into structured signal instead of noise.

Email source	Extracted	Action taken
LinkedIn job alerts	Title, company, location, URL	Appended to new_jobs.csv; generator queued
LinkedIn notifications	Profile views, accepts, InMail replies	Warm leads flagged in tracker (status = warm)
Naukri job alerts	Role, company, city, apply URL	Appended to new_jobs.csv
Recruiter replies	Subject, sender, company, intent	Classified: interested / rejected / follow-up
Interview invites	Date, time, format	Flagged high-priority in daily digest

One-time Gmail setup: about 15 minutes

- Create a project at console.cloud.google.com
- Enable the Gmail API
- Create OAuth 2.0 credentials, download credentials.json
- Run the auth script once, generates token.json (stored locally)
- All future runs use token.json, no browser needed again

Read-only by design. The script only reads mail. It never sends, deletes, labels, or modifies anything in your inbox.

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ALL STATE IN ONE FILE

The Tracker

Every bit of state lives in `tracker.csv`. No database, no dashboard. Each layer reads and writes here, and you can open it in Sheets at any moment to see the whole pipeline.

Column	Updated by	Values
lead_id	generator	auto hash
full_name / company / job_title	generator	from leads.csv
persona_tag	generator	hr · engineer · manager · cto · ceo
domain_tag	generator	startup · scaleup · enterprise · ...
assets_generated	generator	true / false
outreach_sent_date	you	YYYY-MM-DD, after sending
channel	you	email · dm · both
status	inbox monitor	pending » sent » replied » interview
last_reply_date	inbox monitor	YYYY-MM-DD
reply_intent	inbox monitor	interested · rejected · follow-up
follow_up_due	script	sent_date + 5 days
notes	you	free text

Status flow



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WHAT TO EXPECT

Outreach Math

Volume	Replies	Interviews	Notes
500 personalised touches	25–50	10–25	full pipeline run
150 quality touches	8–15	3–7	minimum viable run
50 sniper applications	5–10	2–5	deep research each
1 warm referral	high	≈1 (= 20 cold)	always prioritise

Send cadence

Day	Best windows (IST)	Why
Tuesday	9–11am · 5–7pm	high mid-week open rates
Wednesday	9–11am · 5–7pm	peak recruiter activity
Thursday	9–11am · 5–7pm	before end-of-week reviews
Monday	avoid mornings	weekend inbox overload
Friday	avoid afternoons	pre-weekend wind-down

PACOS never auto-sends. You batch approved assets on Tue/Wed/Thu mornings by copy-pasting from the output folder.

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INTERVIEWS FIRST

Build Order

Build what produces interviews fastest. The visibility layer is last; it compounds over months, not days.

When	Build	Outcome	Priority
Week 1	Asset Generator (CSV » Claude » output/)	Personalised outreach for every lead	● Critical
Week 1	First 50 leads in leads.csv (manual)	Real input, no scraping	● Critical
Week 2	Gmail Inbox Monitor (read-only)	Auto-detect alerts + replies	● High
Week 2	tracker.csv + follow-up logic	Nothing slips through	● High
Week 3	Daily digest script	One place to see today's actions	● Medium
Week 4+	Visibility layer (manual posts)	Inbound signal compounds	● Later
Post-job	OSS cleanup (README, Docker)	Release with a real success story	● Later

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EVERY FUNCTION PRESERVED

Feature Map

Nothing from the original vision was cut. Only the implementation changed, from fragile scraping to simple, reliable inputs.

Function	Was	Now	Kept
Find AI/ML leads	Sales Nav scraping	LinkedIn search + CSV export	✓
Tailor CV per role	JD scraping + LLM	cv_notes.txt via Claude	✓
Cover letter per company	LangGraph agent	cover_letter.txt via Claude	✓
Cold LinkedIn DM	DM agent	cold_dm.txt via Claude	✓
Cold email	outreach agent	cold_email.txt via Claude	✓
Domain-aware tone	classifier agent	domain_tag + system prompt	✓
Monitor job alerts	Playwright	Gmail API parsing	✓
Track recruiter replies	CRM + LangGraph	Inbox Monitor » tracker	✓
Follow-up scheduling	supervisor agent	follow_up_due + digest	✓
LinkedIn visibility	auto-publish	manual copy-paste	✓
Substack / Medium	auto-publish	manual copy-paste	✓
Per-lead personalisation	Proxycurl + LLM	Claude from CSV context	✓
Content dedup	memory store	simple theme log	✓
Application tracker	React dashboard	tracker.csv in Sheets	✓

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WHEN IT'S WORKING

Open Source

Once v1 is running and the job is landed, PACOS is ready to publish, with the most valuable asset any OSS project can have: a genuine “I used this to get hired” story.

File	What it must say
README.md	Problem » solution » one-command setup » screenshot » APIs needed
.env.example	CLAUDE_API_KEY · GMAIL_CREDENTIALS_PATH · LEADS_CSV_PATH
CONTRIBUTING.md	Architecture diagram; how to add a job-board parser or asset type
sample_leads.csv	Five fake rows covering every column for safe testing
Legal note	All lead data is user-supplied; no scrapers shipped; users own ToS compliance
setup script	pip install -r requirements.txt + python auth_gmail.py

Name options

Name	Angle
PACOS	Keep it. The personal origin is a differentiator
ApplyOS	Says exactly what it does
HireLoop	Closed-loop job search
Outbound	Emphasises the outreach-first design

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